probabilities, which are also outside of the acceptable range. So, at first sight, string theory appeared to be awash in its own quantum-mechanical hot water.

With stubborn determination, physicists sought and found the cause of this unacceptable feature. The explanation begins with a simple observation. If a string is constrained to lie on a two-dimensional surface—such as the surface of a table or a garden hose—the number of independent directions in which it can vibrate is reduced to two: the left-right and back-forth dimensions along the surface. Any vibrational pattern that remains on the surface involves some combination of vibrations in these two directions. Correspondingly, we see that this also means that a string in Flatland, the Garden-hose universe, or in any other two-dimensional universe, is also constrained to vibrate in a total of two independent spatial directions. If, however, the string is allowed to leave the surface, the number of independent vibrational directions increases to three, since the string then can also oscillate in the up-down direction. Equivalently, in a universe with three spatial dimensions, a string can vibrate in three independent directions. Although it gets harder to envision, the pattern continues: In a universe with ever more spatial dimensions, there are ever more independent directions in which it can vibrate.

We emphasize this fact of string vibrations because physicists found that the troublesome calculations were highly sensitive to the number of independent directions in which a string can vibrate. The negative probabilities arose from a mismatch between what the theory required and what reality seemed to impose: The calculations showed that if strings could vibrate in nine independent spatial directions, all of the negative probabilities would cancel out. Well, that’s great in theory, but so what? If string theory is meant to describe our world with three spatial dimensions, we still seem to be in trouble.

But are we? Taking a more than half-century-old lead, we see that Kaluza and Klein provide a loophole. Since strings are so small, not only can they vibrate in large, extended dimensions, they can also vibrate in ones that are tiny and curled up. And so we can meet the nine-space-dimension requirement of string theory in our universe, by assuming—à la Kaluza and Klein—that in addition to our familiar three extended spatial dimensions there are six other curled-up spatial